College of Engineering and Technology

Project Work – Student Hand Book

Project Batch ID 255

Degree/ program	B.Tech	**Specialisation**	Computer Science & Engineering
Academic Year	2024-2025 (Even)	**Semester**	6

Name of student	**Register Number**	**Department**	**Mobile Number**	**Email ID**
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Working Title of the Project:	**Hand Gesture-Based Communication System for Disabled Individual**
Project Site / Location	
Name and address of the company / organisation (Applicable for projects with industry or industry support)	SRM IST, Kattankulathur, Chengalpattu District-603203

Supervision Team

	Supervisor	**Co-Supervisor**	**External Supervisor (If applicable)**
Name	Dr. R.Babu		
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SRM INSTITUTE OF SCIENCE & TECHNOLOGY (Deemed to be University u/s 3 of UGC Act, 1956)

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Course Code	21CSP302L	Course Title	Project

Mission Statement

Problem (or) Product Description:
Communication barriers between hearing-impaired individuals and the rest of society continue to create exclusion and misunderstandings, often limiting access to essential services. Traditional solutions like interpreters or written communication are not always practical during everyday interactions. To address this gap, this project proposes a real-time sign language interpretation system built on affordable, portable hardware — specifically the Raspberry Pi 5. It leverages MediaPipe Hands for accurate hand landmark detection, combined with an LSTM-based deep learning model to recognize both dynamic and static signs with high accuracy.
Assumptions and Constraints
Pi 5 Computational power was over estimated
Stakeholders
Disabled Individuals

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Division of work and contributors of SPRINT 1 [Include Daily Scrum of Sprint 1]

Week 1: Project Initiation & Hardware Foundation

Objective:

To establish the project's scope and prepare the foundational hardware stack that will enable gesture tracking and processing on embedded devices.

Key Tasks:

- Defined the problem statement, target users (speech/hearing-impaired individuals), and the project scope aligned with SDG Goals (4 & 10).
- Drafted a requirement specification document listing both hardware (Raspberry Pi 5, Camera Module 3, optional Arduino, Power Bank) and software dependencies (MediaPipe, TensorFlow Lite, SQLite, Flask).
- Evaluated multiple camera modules and finalized the Raspberry Pi Camera Module 3 for its latency and resolution performance.
- Conducted initial tests:
  - Verified Pi-camera connection.
  - Tested streaming & frame capture using OpenCV.
  - Assessed Power Bank output stability for portability.

Outcome:

A clear development roadmap, finalized hardware stack, and basic hardware testing environment — ready for the next phase of gesture system development.

Signature of the Supervisor

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Division of work and contributors of SPRINT 2 [Include Daily Scrum of SPRINT 2]

Objective:

Build the gesture detection system and prepare a model-ready data pipeline using landmark detection.

Key Tasks:

- Integrated camera feed into a Python script to capture real-time hand movements.
- Deployed MediaPipe Hands to extract 21 hand landmarks per frame, supporting both static and dynamic gestures.
- Normalized coordinates and removed jitter using low-pass filters for stable input signals.
- Designed a gesture-to-label classification model:
  - Chose LSTM for dynamic temporal gesture patterns.
  - Collected samples of signs like "Help", "Water", "Stop" from test users for training data.
- Stored gestures with labels into SQLite for future retrieval & updates.

Outcome:

Successfully built and tested the hand gesture tracking engine; extracted clean landmark features; defined input schema for ML models.

9/05/25

Signature of the Supervisor

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Division of work and contributors of SPRINT 3 [Include Daily Scrum of Sprint 3]

Week 3: NLP Integration, TTS & Real-Time Sentence Generation

Objective:

Enhance communication output by converting gesture predictions into full sentences using NLP + voice output systems.

🔧 **Key Tasks:**

- **Used Gemini AI to interpret recognized keywords (e.g., "I", "need", "help") into complete grammatical sentences like "I need help urgently."**
- **Integrated PicoTTS (SVOX Pico) engine for offline, low-latency voice synthesis.**
- **Built a Flask-based web interface hosted on Raspberry Pi:**
  - **Displayed live predictions and built sentences.**
  - **Allowed manual corrections by the user or caregiver.**
  - **Enabled customization of pre-defined gesture responses.**
- **Implemented Bluetooth A2DP output to stream synthesized voice to wireless speakers/headsets.**
- **Developed feedback mechanisms (LED on gesture recognition, optional haptic trigger for gestures) for visually-impaired users or private use.**

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- Developed feedback mechanisms (LED on gesture recognition, optional haptic trigger for gestures) for visually-impaired users or private use.

Outcome:

Full gesture → keyword → sentence → audio pipeline complete, with both real-time display and sound feedback. Multi-modal communication output enabled.

Signature of the Supervisor